

Mathematics for Biomedical Engineering

L7:

Complex Numbers

Dr. Michele Orini
King's College London

November 19th, 2025



- To understand the basic definition of complex numbers;
- To perform simple algebraic tasks with complex numbers;
- To understand the different representations of complex numbers (standard, polar, trigonometric);
- To be able to represent and interpret a complex number on an Argand diagram;
- To be able to solve complex equations involving complex numbers;
- To understand what a phasor is and use phasors to study phase relationships in a.c. circuits.

- 1 Arithmetic of Complex Numbers
- 2 The Argand Diagram
- 3 The Exponential (Polar) Form of Complex Numbers
- 4 De Moivre's Theorem
- 5 Solving Equations and Finding Roots of Complex Numbers
- 6 Phasors

The square root of a negative number

The study of **complex numbers** enables the set of *real numbers* to be extended. Complex numbers are useful in a variety of applications, particularly alternating current (see Electrical Engineering course next Semester).

Complex numbers relies on the introduction of an imaginary number, i .

Imaginary Number

i is an **imaginary number** such that

$$i^2 = -1 \qquad i = \sqrt{-1} \qquad (1)$$

Example: Find $\sqrt{-9}$

Example: Use the formula for a quadratic equation to solve $2x^2 + x + 1 = 0$

Complex numbers

A **complex number** is made-up of two parts: a **real part** and an **imaginary part**. The imaginary part is the multiple of i .

A Complex Number

If z is a complex number

$$z = a + ib \quad (2)$$

where a is the real part and b is the imaginary part.

The real part is often written as $Re(z)$ and the imaginary part as $Im(z)$.

Two complex numbers are equal *only* when *both* their real *and* their imaginary parts are also equal i.e. if $z_1 = z_2$, where $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, then $a_1 = a_2$ and $b_1 = b_2$.

Complex number magnitude

The magnitude of a complex number is denoted by $|z|$ and given by

$$|z| = \sqrt{a^2 + b^2} \quad (3)$$

Complex conjugates

A **complex conjugate** of a complex number involves switching the sign of the imaginary part of the complex number; *i.e.*, switching $+ib \leftrightarrow -ib$.

Complex Conjugate

If $z = a + ib$, then the **complex conjugate** of z is denoted by $\bar{z}$ (or z^*) and given by

$$\bar{z} = a - ib \quad (4)$$

Note, that the magnitude of a complex conjugate is the same as the magnitude of the complex number

Complex conjugate magnitude

$$|\bar{z}| = \sqrt{a^2 + b^2} = |z| \quad (5)$$

Example: What are the complex conjugates of $z_1 = 11 + 2i$, $z_2 = 8$, $z_3 = -9i$?

Addition & subtraction of complex numbers

We find the sum and difference of complex numbers by adding/subtracting the real and imaginary parts separately.

Addition and subtraction of complex numbers

If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ then

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2)i \quad (6)$$

$$z_1 - z_2 = (a_1 - a_2) + (b_1 - b_2)i \quad (7)$$

Multiplication of complex numbers

To multiply two complex numbers, the normal rules of algebra apply, but remembering that $i^2 = -1$.

Multiplying complex numbers

If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ then

$$z_1 z_2 = (a_1 + ib_1)(a_2 + ib_2) \quad (8)$$

$$= a_1 a_2 + a_1 b_2 i + b_1 a_2 i + b_1 b_2 i^2 \quad (9)$$

$$= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + b_1 a_2)i \quad (10)$$

A very important result of complex number multiplication is that the product of a complex number with its conjugate is equal to the magnitude of the complex number squared

Complex multiplied by its conjugate

$$z \bar{z} = (a + ib)(a - ib) = a^2 + b^2 = |z|^2 \quad (11)$$

Division of complex numbers

Division of complex numbers is a bit more tricky and requires a special technique: **multiply both numerator and denominator by the complex conjugate of the denominator.**

Dividing complex numbers

$$\frac{z_1}{z_2} = \frac{a_1 + b_1 i}{a_2 + b_2 i} \quad (12)$$

$$= \frac{a_1 + b_1 i}{a_2 + b_2 i} \times \frac{a_2 - b_2 i}{a_2 - b_2 i} \quad (13)$$

$$= \frac{a_1 a_2 + b_1 b_2 + (a_2 b_1 - a_1 b_2)i}{a_2^2 + b_2^2} \quad (14)$$

Remember that $z\bar{z} = |z|^2 = a^2 + b^2$.

Example: If $z_1 = 2 + 9i$ and $z_2 = 5 - 2i$, find $\frac{z_1}{z_2}$.

Example: Simplify $\frac{3}{-i}$.

Representing complex numbers on an Argand Diagram

An **Argand Diagram** provides a way of representing the complex number $z = a + ib$ in a 2D plot as a point with coordinates (a, b) .

The real part of z is plotted on the horizontal axis (the **real axis**), and the imaginary part of z plotted on the vertical axis (the **imaginary axis**).

We refer to this diagram as the **complex plane**.

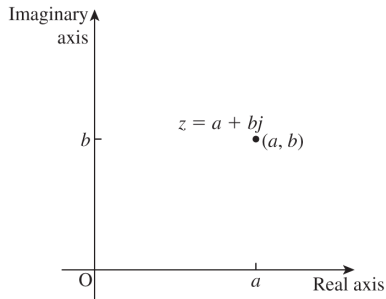


Figure: Representation of the complex number $z = a + ib$ in an Argand Diagram

Example: Plot the complex numbers $2 + 3i$ and $-3 - 2i$ in the complex plane.

The modulus of a complex number

Modulus

The magnitude, or **modulus**, of a complex number $z = a + ib$ is represented by

$$|z| = \sqrt{a^2 + b^2} \quad (15)$$

In the complex plane, the modulus represents the distance from the origin, r .

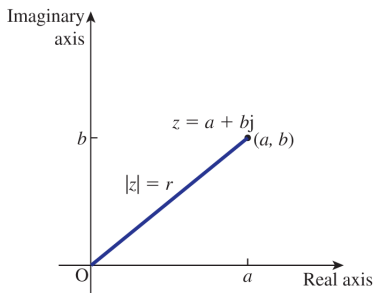


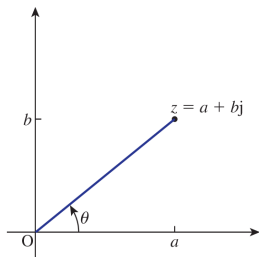
Figure: Modulus of a complex number z in the complex plane as the distance of (a, b) from O .

The argument of a complex number

Argument

The **argument** of a complex number $z = a + ib$ is the angle between the positive x-axis and the line joining (a, b) to the origin, and is usually signified by θ

$$\theta = \arg(z) = \tan^{-1} \left(\frac{b}{a} \right) \quad (16)$$



Be careful when using a calculator!

Always plot the complex number in an Argand Diagram to check!

Figure: Argument of z in the complex plane.

Example: Find the arguments of $1 + \sqrt{3}i$, $1 - \sqrt{3}i$, $-1 + \sqrt{3}i$ and $-1 - \sqrt{3}i$.

Polar form

A complex number (and thus its location in the complex plane) can be uniquely specified by its modulus and argument - known as **Polar Form**.

Polar form

The **polar form** of a complex number can be written as

$$z = r\angle\theta \tag{17}$$

where r is the modulus and θ is the argument.

The form $r(\cos \theta + i \sin \theta)$

From trigonometry, we can easily see that $\cos \theta = \frac{a}{r}$ and $\sin \theta = \frac{b}{r}$.

This can be rearranged to give the real (a) and imaginary (b) parts of z .

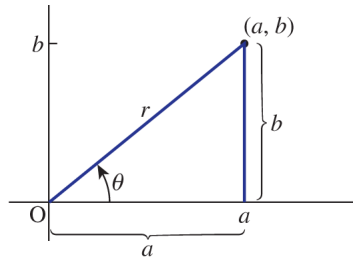


Figure: Relating r and θ to a and b of a complex number

Trigonometric form of a complex number

We can also write a complex number $z = a + ib$ as

$$z = r(\cos \theta + i \sin \theta) \quad (18)$$

Example: Express $z = 4\angle(\frac{\pi}{3})$ in standard form.

The conjugate form $r(\cos \theta + i \sin \theta)$

Note that $\bar{z}$ has the same real part but the sign of the imaginary part is switched:

$$\bar{z} = r(\cos \theta - i \sin \theta)$$

This is because $\cos(-\theta) = \cos \theta$, whereas $\sin(-\theta) = -\sin \theta$.

This represents the reflection in the real axis in the complex plane.

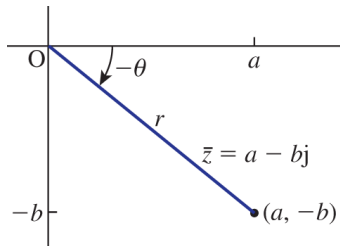


Figure: Complex conjugate in the complex plane.

Multiplication & division of complex numbers in polar form

Using polar form makes multiplying and dividing complex numbers very simple.

- **Multiplying:** multiply the moduli, add the arguments;
- **Dividing:** divide the moduli, subtract the arguments.

Multiplying & dividing complex numbers

If $z_1 = r_1 \angle \theta_1$ and $z_2 = r_2 \angle \theta_2$ then

$$z_1 z_2 = r_1 r_2 \angle (\theta_1 + \theta_2), \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2) \quad (19)$$

Example: If $z_1 = 5 \angle (\frac{\pi}{6})$ and $z_2 = 4 \angle (\frac{-\pi}{4})$, find $z_1 z_2$ and $\frac{z_1}{z_2}$.

- 1 Multiplying a complex number by i causes an anticlockwise rotation by $\frac{\pi}{2}$ on the Argand Diagram.
- 2 Dividing a complex number by i causes a clockwise rotation by $\frac{\pi}{2}$ on the Argand Diagram.
- 3 If $z = r \angle \theta$, then $z^n = r^n \angle n\theta$ (De Moivre's Theorem - see later).

Euler's relations

The power series expansions of e^x , $\sin x$ and $\cos x$ can be written as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad (20)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \quad (21)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \quad (22)$$

Using these we can derive expressions for $e^{i\theta}$, $\sin \theta$ and $\cos \theta$.

Combining these relations gives **Euler's Relations**

Euler's Relations

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (23)$$

$$e^{-i\theta} = \cos \theta - i \sin \theta \quad (24)$$

[Remembering that $\cos(-\theta) = \cos \theta$, whereas $\sin(-\theta) = -\sin \theta$.]

The exponential form of a complex number

Using Euler's relations $e^{i\theta} = \cos \theta + i \sin \theta$ and $e^{-i\theta} = \cos \theta - i \sin \theta$, adding and then subtracting gives expressions that can be re-arranged to give:

Expressions for cos and sin

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \qquad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \qquad (25)$$

Example: Show that $3 \left(\frac{e^{i\omega} - e^{-i\omega}}{i\omega} \right) = \frac{6 \sin \omega}{\omega}$

The exponential form of a complex number

If also follows immediately from Euler's relations that

Exponential form of a complex number

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta} \quad (26)$$

$$\bar{z} = r(\cos \theta - i \sin \theta) = re^{-i\theta} \quad (27)$$

Example: Express the complex number $z = 3 + 3i$ in exponential form.

Example: In control theory, complex numbers of the form $z = e^{st}$ are used, where $s = \sigma + i\omega$, where ω is frequency, t is time and σ is a real number. Find the real and imaginary parts of z and its modulus.

De Moivre's Theorem

By raising the complex number $z = e^{i\theta}$ to the power n , we can see that

$$z^n = (e^{i\theta})^n = e^{in\theta}. \quad (28)$$

However, this also applies if the modulus $r \neq 1$ (i.e. if $z = re^{i\theta}$) because

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} \quad (29)$$

which gives:

De Moivre's Theorem

$$z^n = (r(\cos \theta + i \sin \theta))^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta) \quad (30)$$

De Moivre's Theorem can also be used in polar form (as we stated before!)

$$z^n = r^n \angle n\theta \quad (31)$$

Example: If $z = 7 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$, find z^2 .

Using De Moivre's Theorem

De Moivre's Theorem can be used to prove trigonometrical identities.

For example, use De Moivre's Theorem to prove that

$$\sin 2\theta = 2 \sin \theta \cos \theta \qquad \cos 2\theta = \cos^2 \theta - \sin^2 \theta \qquad (32)$$

Using the fact that

$$z^n = \cos n\theta + i \sin n\theta \qquad (33)$$

$$z^{-n} = \cos n\theta - i \sin n\theta \qquad (34)$$

we can separately add and subtract these two expressions to derive that

De Moivre relationships

If $z = \cos \theta + i \sin \theta$ then

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \qquad (35)$$

$$z^n - \frac{1}{z^n} = 2i \sin n\theta \qquad (36)$$

Example: Show that $\cos^2 \theta = \frac{1}{2} \cos 2\theta + \frac{1}{2}$.

Solving equations

When solving a complex number equation **of degree** n we always require n solutions.

For example, the equations

$$z^3 + 5z = 0$$

requires 3 solutions, which we can show are $z = 0, z = -\sqrt{5}i, z = \sqrt{5}i$.

For the equation

$$z^2 + 2z + 2 = 0$$

we require 2 solutions and these can be found by applying the quadratic formula to be $z = -1 \pm i$.

Complex roots & the Argand Diagram

Often it is more convenient to work in polar or exponential form.

Important concept regarding Arguments of complex numbers

If we add on 360° , or 2π (or indeed any multiple thereof), to the argument of a complex number, we remain at exactly the *same point within the complex plane*, and thus the complex number is *unchanged*!

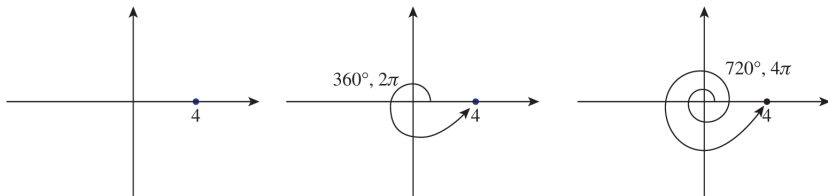


Figure: The argument of a complex number ($z = 4$) is unchanged by the addition of $2\pi, 4\pi, \dots$

In polar, or exponential form, we may write this as $4\angle 0$, $4\angle 2\pi$, $4\angle 4\pi$ (or $4e^{i0}$, $4e^{i2\pi}$), $4e^{i4\pi}$. Note that we often write this as $4\angle 2k\pi$ or $4e^{i2n\pi}$

Example: Express $z = 2 + i$ in polar form including the general form of the argument.

Solving complex equations using polar / exponential form

To solve a complex number equations such as

$$z^3 = 4$$

follow the following steps:

- 1 Write down the RHS of the equation in polar/ exponential form **with a general argument**.

$$4 = 4e^{i(0+2k\pi)}$$

- 2 Equate the LHS to the RHS

$$z^3 = 4e^{i(2k\pi)}$$

- 3 Take the root of both sides of the equations (depending on the order of the equation)

$$z = \sqrt[3]{4e^{i(2k\pi)}} = 4^{1/3} e^{i2k\pi/3}$$

- 4 Cycle through values of k , starting from 0, until you have found all roots (i.e. $k = 0 \rightarrow n - 1$)

$$k = 0 \Rightarrow z_0 = 4^{1/3}, \quad k = 1 \Rightarrow z_1 = 4^{1/3} e^{i2\pi/3}, \quad k = 2 \Rightarrow z_2 = 4^{1/3} e^{i4\pi/3}$$

- 5 Plot the roots on the Argand Diagram and check they are evenly spaced (and that you have n of them!)

Examples of complex equations

Example: Solve $z^2 = 2 + i$.

Example: Find the roots of $(z - 3i)^4 = 3i$.

Phase relationships in circuits

One of the most important applications of complex numbers in engineering is in the study of *alternating current* (a.c.) circuits.

The phase relationship between different components in a circuit can be expressed using complex numbers.

A **phasor** is a quantity that contains information about the peak voltage or peak current in a component together with its phase.

The current in an a.c. circuit can be expressed in the form

Voltage in an a.c. circuit

$$v(t) = V \sin(\omega t + \phi) \quad (37)$$

where V is the peak voltage, ω the angular frequency, ϕ the phase relative to some reference voltage and t is time.

Similarly, current in the circuit can be expressed as

$$i(t) = I \sin(\omega t + \phi) \quad (38)$$

where I is the peak current.

Resistors, Inductors, Capacitors

If the current in each circuit is $i(t) = I \sin \omega t$, in a circuit containing a:

Resistor (of resistance R), the current is in phase with the voltage

$$v_R(t) = IR \sin \omega t$$

Inductor (of inductance L), the voltage *leads* the current by a phase of $\frac{\pi}{2}$

$$v_L(t) = \omega LI \sin \left(\omega t + \frac{\pi}{2} \right)$$

Capacitor (of capacitance C), the voltage *lags* the current by a phase of $\frac{\pi}{2}$

$$v_C(t) = \frac{I}{\omega C} \sin \left(\omega t - \frac{\pi}{2} \right)$$

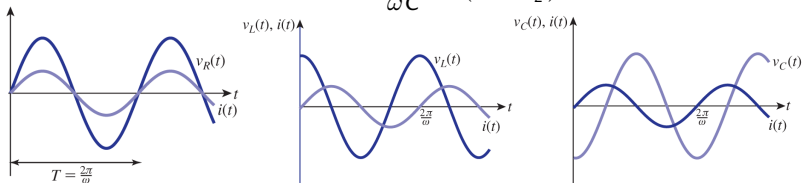


Figure: Current and voltage in a resistor are in phase. Voltage leads current by a phase of $\frac{\pi}{2}$ in an inductor. Voltage lags current by a phase of $\frac{\pi}{2}$ in a capacitor.

Phasors

Consider a voltage waveform $V_1 \sin \omega t$ in an a.c. circuit. The time it takes to complete one cycle is $T = \frac{2\pi}{\omega}$.

An alternative representation is an arm of length V_1 rotating at ω radians per second - such a representation is called a **phasor**.

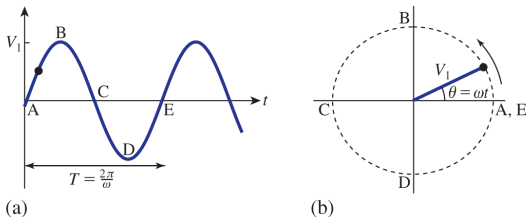


Figure: Points on a waveform and equivalent points on a rotating arm.

We can think of the phasor diagram as an Argand Diagram, and thus write a phasor in *polar form*

Phasor as a complex number

$$\tilde{V} = V \angle \phi = V e^{i\phi} \quad (39)$$

Phasors

If we have a second waveform $V_2 \sin(\omega t + \frac{\pi}{2})$ which leads V_1 by $\frac{\pi}{2}$, then the rotating arm will also lead that of V_1 by $\frac{\pi}{2}$.

Similarly, if we have a third waveform $V_3 \sin(\omega t - \frac{\pi}{2})$ which lags V_1 by $\frac{\pi}{2}$, then the rotating arm will also lag that of V_1 by $\frac{\pi}{2}$.

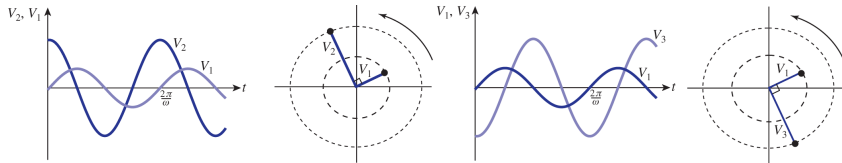


Figure: Points on a waveform and equivalent points on a rotating arm for leading (left) and lagging (right) cases.

Recall that we could rotate a complex number by $\frac{\pi}{2}$ on an Argand Diagram by multiplying it by i .

Conversely, we can rotate a complex number by $-\frac{\pi}{2}$ on an Argand Diagram by multiplying it by $-i$ (or dividing by i).

Phasors

Thus, if the current in a circuit is represented by $\tilde{I}$, the voltage across:

Resistor will be $\tilde{V}_R = RI$

Inductor will be $\tilde{V}_L = i\omega L\tilde{I}$

Capacitor will be $\tilde{V}_C = -\frac{i}{\omega C}\tilde{I}$

which introduces the required phase shifts.

Complex Impedances

$$\tilde{V} = \tilde{I}Z \quad \text{where } Z = \begin{cases} R & \text{for a resistor} \\ i\omega L & \text{for an inductor} \\ -\frac{i}{\omega C} & \text{for a capacitor} \end{cases}$$

Other applications of complex numbers

- ① **Fourier Transforms** - any periodic function can be expressed as the sum of a series of sines and cosines (of varying amplitudes)

$$F(k) = \int_{-\infty}^{\infty} f(x)e^{-i2\pi kx} dx$$

- ② **k-space in MR** - important applications of complex numbers and Fourier Series.
- ③ **The Wave Equation** - information about the speed of travel and frequency of oscillation of a wave (e.g., in optics) can be represented by

$$A = A_0 e^{i(kx - \omega t)}$$

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- To understand what a phasor is and use phasors to study phase relationships in a.c. circuits.